Navnith Bharadwaj

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EDUCATION

University of Colorado Boulder – Aug 2025 – May 2027
Master of Science in Computer Science – GPA 3.74 / 4.00 – Boulder, Colorado

- **Relevant Coursework:** CSCI 5922 Neural Nets and Deep Learning, CSCI 5722 Computer Vision, CSCI 5832 Natural Language Processing, CSCI 5214 Big Data Architecture, CSCI 5244 Quantum Computation

R V College of Engineering – Aug 2019 – Jul 2023
Bachelor of Engineering in Computer Science and Engineering – Bengaluru, India

- Relevant Coursework: Data Structures, Algorithms, Operating Systems, Software Engineering, DBMS, OOP

TECHNICAL SKILLS

Languages: Python, **TypeScript**, Java, C++, JavaScript, SQL, Bash
Backend and APIs: FastAPI, Spring Boot, Node.js, REST, GraphQL, gRPC, **microservices**, WebSockets, OAuth, **React**
Machine Learning and AI: PyTorch, TensorFlow, LLM APIs, **RAG**, **agentic pipelines**, **distributed training**, computer vision, model evaluation
Cloud and Infrastructure: Kubernetes, Docker, GKE, AWS, Azure, GCP, CI/CD, Ansible, Git, Linux, **AI-assisted development**
Databases and Tools: PostgreSQL, MySQL, MongoDB, Redis, **OpenCV**, scikit-learn, pytest, code reviews, **latency profiling**, debugging

EXPERIENCE

AI Software Engineer Intern – Aug 2026 – Present
Lyntra – Remote

- Built **consent-scoped cohort messaging** with mandatory **idempotency keys** and deterministic uuid5 fingerprints over a **frozen-audience batch row**, so a retry after a lost response redelivers what was authored rather than the current roster.
- **Separated persistence from delivery** across 3 channels, treating a 2xx as delivered only when an **in-app receipt** exists, and **enforcing consent in SQL** so revoking a relationship retracts messages already in a student's inbox.
- Built **3-layer deduplication** for Sheets sync using a normalizer that unifies openpyxl and Sheets API cell types, a **content hash** over the normalized grid, and a **file-keyed lineage key** so edited rows update in place instead of duplicating.
- Gated every model call behind a **fail-closed allowlist** so tutor workbooks carrying parent phone numbers and portal logins **never reached an LLM**.

Software Developer – Aug 2023 – Aug 2025
Oracle – Bengaluru, India

- Led an **internal developer platform** for **Backup and Disaster Recovery** spanning VM backup, database backup and recovery, and VM DR end-to-end, directing 1 engineer, adopted by **3 infra teams**.
- Designed and shipped 35 **RBAC-secured REST APIs** and **GraphQL** schemas standardizing backup and recovery operations, cutting average request-to-execution time from **6 hours to 5 minutes**.
- Built **asynchronous execution pipelines** for long-running backup and recovery jobs, cutting user-facing wait times by **45%** for **high-volume data processing**.
- Instituted **CI/CD** pipelines, **code reviews**, and unit and integration tests at **85% coverage**, cutting production rollback incidents by **35%**.

Backend Developer Intern – Sept 2022 – Nov 2022
Ideoholics – Remote

- Shipped **Java** and Spring Boot REST APIs within a microservices architecture, sustaining **200,000 requests per day** at **99.5%** uptime.

PROJECTS

MeshML: Distributed Platform for ML Training | Python, PyTorch, Kubernetes, gRPC, Docker

- Built a **parameter-server architecture** over 6 gRPC services with bidirectional streaming and **bounded-staleness gradient aggregation**, keeping **async data-parallel SGD** convergent across 4 heterogeneous CPU and GPU workers.
- Orchestrated training on **GKE Kubernetes** with horizontal autoscaling, 256-sample **dataset sharding** and **object-storage checkpointing** so a lost worker resumes instead of restarting the run.

ML Model A/B Testing Framework | Python, FastAPI, scikit-learn, PostgreSQL

- Engineered a **FastAPI serving layer** to deploy and evaluate model versions against live traffic, scoring accuracy, F1, AUC and **p95/p99 latency** with **Welch's t-test** to gate releases.

F1 Telemetry-to-Narrative: AI Agent Pipeline | Python, MLX, OpenAI, Pydantic

- Co-built a 6-stage **agentic AI pipeline** over OpenAI and on-device MLX with **RAG-style retrieval** and Pydantic-validated output.

Fire Detection on Drone-Captured Images | Python, YOLOv8, OpenCV

- Co-built drone fire detection, training **YOLOv8n, s and m** on synthetic and real datasets and ensembling to **0.99 precision**.

Fingerprint Ridge-Wear Analysis for Occupational Inference | Python, scikit-learn, scikit-image, OpenCV

- Tested whether fingerprint ridge-wear predicts occupation over 248 images, reporting a **negative result** after a **confound check**.